

Visual Decoding from EEG

A Project Report Submitted by

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in partial fulfilment of the requirements for the award of the degree of

BTech in Data Science and Engineering



Indian Institute of Technology Mandi
School of Computing and Electrical Engineering

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Declaration

I hereby declare that the work presented in this Project Report titled Visual Decoding from EEG submitted to the Indian Institute of Technology Mandi in partial fulfilment of the requirements for the award of the degree of BTech in Data Science and Engineering, is a bonafide record of the research work carried out under the supervision of Prof.Arn timer Bhaskar and Prof.Padmanabhan. The contents of this Project Report in full or in parts, have not been submitted to, and will not be submitted by me to, any other Institute or University in India or abroad for the award of any degree or diploma.

Signature

Atharv Kumar (B21038)

Certificate

This is to certify that the Project Report titled Visual Decoding from EEG, submitted by Atharv Kumar(B21038) to the Indian Institute of Technology Mandi for the award of the degree of BTech in Data Science and Engineering, is a bonafide record of the research work done by him under my supervision. To the best of my knowledge, the contents of this report, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

Signature

Prof.Arn timer Bhaskar

Prof.Padmanabhan

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I would like to extend my gratitude to all those who contributed to the successful completion of this project "Visual Decoding from EEG."

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I would also like to acknowledge the authors of the base papers which we have followed "DreamDiffusion: Generating High-Quality Images from Brain EEG Signals" whose work and methodology inspired us in many aspects of this project. Their contributions laid the base for exploring EEG-to-image reconstruction tasks.

Lastly I am very much thankful to our friends for their support and encouragement throughout the project.

Thank you all for the successful completion of the project !!

Abstract

The project tries to solve the challenging task of visual decoding from EEG signals, aiming to reconstruct images as perceived by the brain using EEG (with high temporal resolution) and deep learning techniques. We have worked on a dataset which comprise of EEG Signals, corresponding images and labels. We have developed a comprehensive pipeline. EEG signals were encoded into embeddings using a Variational Auto-Encoder (VAE). Image features were extracted in two forms: depth maps obtained via GCNN (a graph neural network based architecture consisting of GAT/GCN layers) and textual descriptions generated using the CLIP model(integrated with GPT-2 Decoder). Note the Ground truth Text for the corresponding EEG embeddings is been given by BLIP-2 the present state of the art Image Captioning model. Further the generated text through CLIP using Cross-Modal Alignment and GPT-2 decoder and the Depth map through our trained GCNN network architecture is been sent to the pre-trained stable diffusion model where Depth and Text information is used for reconstruction of images.Thus, we get the reconstructed image directly from the EEG signal contributing to the Visual Decoding using EEG framework or problem. We have used metrics like cosine similarity to check the quality of the optimized embeddings and SSIM score to check the quality of the reconstructed image.

The proposed framework not only reconstructs images from EEG signals, but also provide insights into how different modalities like textual prompts, depth maps and EEG embeddings contribute to visual decoding.

We conducted experiments with a pre-trained Stable Diffusion model to test different combinations of inputs, including noisy images, depth maps, and textual prompts. Results revealed that textual descriptions from BLIP-2 and CLIP served as the most effective prompt for image reconstruction. Finally, the generated text through CLIP using Cross-Modal Alignment and GPT-2 decoder and the Depth map through our trained GCNN network architecture is been sent to the pre-trained stable diffusion model where Depth and Text information is used for reconstruction of images. This methodology demonstrates a novel approach to decoding visual information from neural activity, potentially advancing applications in neuroscience and assistive technologies.

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Visual Decoding from EEG

1 Introduction and background

The human brain encodes a very large amount of information about the world, and decoding this information will give insights into cognitive processes and neural representations. Among various brain-computer interface technologies, electroencephalography (EEG) has emerged as a practical tool due to its non-invasive nature and temporal resolution. EEG signals represent brain activity in real-time, making them a valuable resource for exploring how humans perceive, process, and interpret visual information. However, translating EEG signals into interpretable formats, such as reconstructed images, remains a challenging problem.

Some recent works, such as MinD-Vis attempt to reconstruct visual information based on fMRI (functional Magnetic Resonance Imaging) signals, which is another way to measure brain activities. They have demonstrated the feasibility of reconstructing high-quality results from brain activities. However, they are still far away from our goal of using brain signals to create conveniently and efficiently. 1) Since fMRI equipment is not portable and needs to be operated by professionals, it is difficult to capture fMRI signals. 2) The cost of fMRI acquisition is high. They greatly hinder the widespread use of this method in the practical artistic generation. In contrast, EEG (electroencephalogram) is a non-invasive and low-cost method of recording electrical activity in the brain. Portable commercial products are now available for the convenient acquisition of EEG signals, showing great potential for future art generation.

This project "Visual Decoding from EEG" aims to bridge the gap between neural signals and visual representations. Leveraging the advanced deep learning models like Variational Autoencoders (VAE), GCNN networks, BLIP-2, CLIP and Stable Diffusion, this work explores an innovative approach to reconstructing images solely from EEG data. The methodology combines embeddings extracted from EEG, textual descriptions and depth maps. We have created a comprehensive pipelines for cross-modal alignment and image reconstruction. This project have potential applications in neuroscience and human-computer interaction.

Background

EEG-based image reconstruction has gained significant attention in recent years, driven by advancements in machine learning and multimodal integration.

Traditional methods rely on fMRI-image paired data to train the model to predict image features from fMRI. These images are fed into GANs for stimulus reconstruction during testing. However recent studies have proposed unsupervised approaches, such as a reconfigurable autoencoder designs. The drawback of using the fMRI signals is that fMRI equipment is not portable and needs to be operated by professionals. Secondly cost of fMRI acquisition is high and also they have high spatial resolution but low temporal resolution.

Generating images from EEG has also been explored using deep learning techniques like Brain2Image,

ThoughtViz etc. Overall these approaches demonstrate the potential of using brain signals to generate images and advance the field of brain-computer interfaces.

State-of-the-art methods, such as Dream Diffusion and BrainVis, have demonstrated that aligning EEG embeddings with visual or textual features can facilitate image generation. For instance, guided diffusion models have shown the ability to map neural signals to both coarse and fine-grained image features, while text-image alignment frameworks like CLIP enable semantic understanding of EEG embeddings. Despite these advances, challenges remain in capturing the rich temporal and frequency-domain information of EEG signals and aligning them effectively with visual features

This project builds on the foundational ideas of EEG-based multimodal learning, integrating insights from recent research. The use of BLIP-2, CLIP and GPT-2 decoder in an integrated manner for generating textual captions from images and GCNN network for extracting depth information represents innovative steps toward improving reconstruction fidelity. Additionally, leveraging the capabilities of Stable Diffusion, a powerful generative model, offers the potential to decode not only semantic but also stylistic aspects of visual stimuli.

We seek to push the boundaries of visual decoding from brain activity by combining EEG embeddings with textual and depth map features. The project tries to offer a robust framework for reconstructing images from neural signals with improved accuracy.

2 Problem definition and Objective

The problem of visualizing the brain activity is a long standing challenge in neuro-science and artificial intelligence. Despite significant advances in brain-computer interfaces reconstructing images from non-invasive EEG signals remains difficult due to several inherent limitations like Low Spatial Resolution, High dimensionality, Cross Modal Alignment and Generative complexity. Traditional methods use fMRI in place of EEG signals for brain computer interaction but fMRI faces several issue like 1) fMRI equipment is not portable and needs to be operated by professionals, it is difficult to capture fMRI signals. 2) The cost of fMRI acquisition is high. They greatly hinder the widespread use of this method in the practical artistic generation. In contrast, EEG (electroencephalogram) is a non-invasive and low-cost method of recording electrical activity in the brain. Portable commercial products are now available for the convenient acquisition of EEG signals. Thus using EEG signals for visual decoding gives us a robust and portable solution to brain-computer interfaces.

Objectives

The objectives of our project can be summed down to major three :

Develop a Visual Encoding Pipeline Using EEG Signals: Create a robust framework that can extract meaningful features from EEG signals and use them to represent and understand visual stimuli effectively.

Integrate Multi-Modal Techniques for Image Reconstruction: Leverage image captioning (using BLIP) and image generation (using Stable Diffusion) to reconstruct high-quality images based on brain activity data.

Enable Direct "Thought-to-Image" Conversion: Achieve seamless image generation from EEG signals without relying on textual intermediates, demonstrating the potential of brain activity-driven visual content creation.

3 Methodology

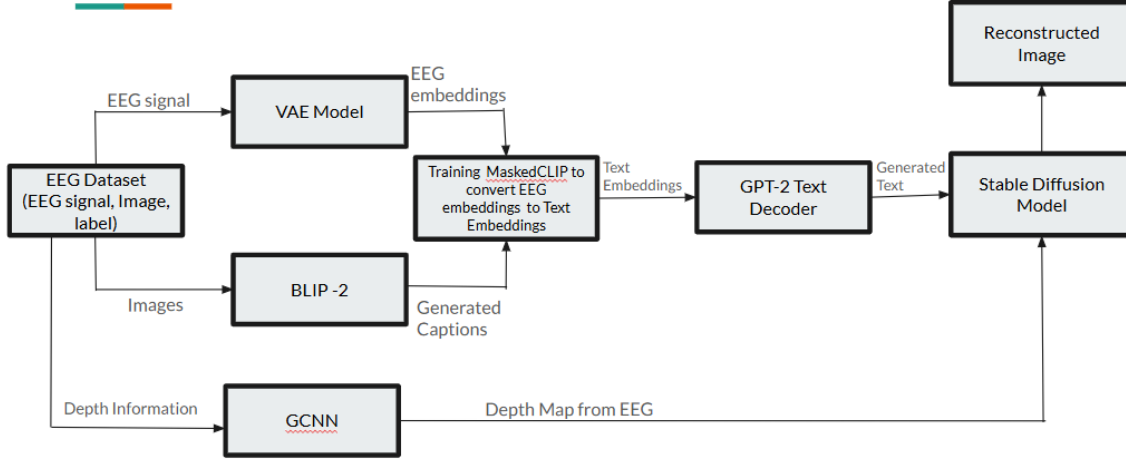


Figure 3.1: Overall Model Followed (note that at testing time we have only EEG signals)

The methodology for the project "Visual Decoding from EEG" involves a multi-modal approach combining EEG signals, depth maps, and textual embeddings to reconstruct images (decode visual stimuli using EEG). The whole process is divided into two organized pipelines one pipeline being extracting depth from the EEG (EEG to Depth) where we bring the depth map embeddings and the EEG embeddings closer and work for EEG to Depth Conversion using GCNN network architecture. The second pipeline involves caption generation from the images and then get the text embeddings and then bring both the text and the EEG embeddings closer and fine tune CLIP model to convert EEG to Text . Later both the pipelines come together by using a projection layer and are fed as a input into a stable diffusion model which produces the reconstructed image and learns to reconstruct image based on modified EEG embeddings. Thus in the testing phase we can take EEG embeddings, modify it and then put forward that modified embeddings in the stable diffusion model to get the reconstructed image, based on the EEG data. (visual decoding using EEG).

The whole process is structured into the following steps:

1. Data Acquisition and Pre-processing

The dataset consists of EEG signals, corresponding visual images, and labels. Each data modality undergoes specific preprocessing:

EEG Signals: Raw EEG data. The structure of the data is as follows : We have 16740 samples, with 17 channels and 100 time-points.

Images: Images corresponding to the EEG signals are resized and normalized to match the input requirements of various models.

Labels: Textual descriptions and categories serve as supplementary data for multi-modal alignment. We have in total 40 labels.

(Note the same Images(16740) were shown to 10 separate subjects and the EEG signal was recorded. This is we have a data of 16740 eeg samples for each of the 10 subjects.)

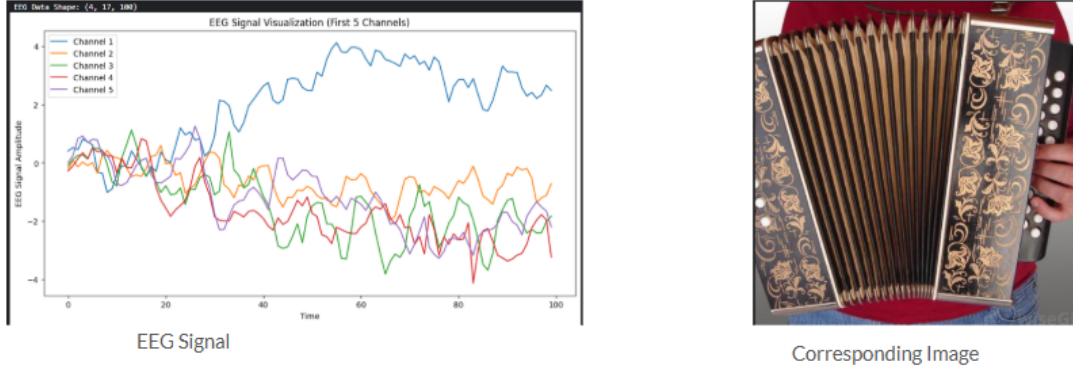


Figure 3.2: EEG data (plotting only 5 out of 128 channels) and its corresponding image

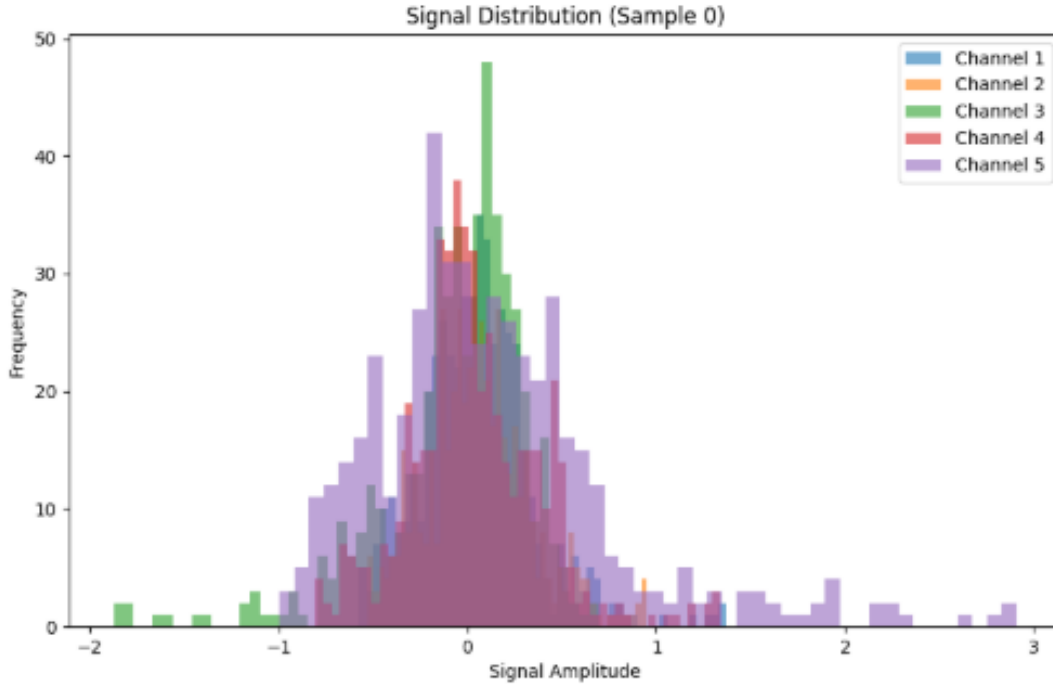


Figure 3.3: Signal Distribution of EEG signal considering 5 out of 128 channels

2. EEG Embedding Extraction

We have used Variational Auto-Encoder to convert the EEG signals into embeddings. A VAE model encodes EEG signals into latent embeddings, capturing essential neural features while reducing dimensionality. This step transforms the temporal and spatial complexity of EEG data into a compact and meaningful representation. We have tried out various models to extract the best embeddings from the EEG signal like LabRAM, hvEEGNet, VQ-VAE etc but it is to our finding though experimentation that

VAE works the best with our pipeline.

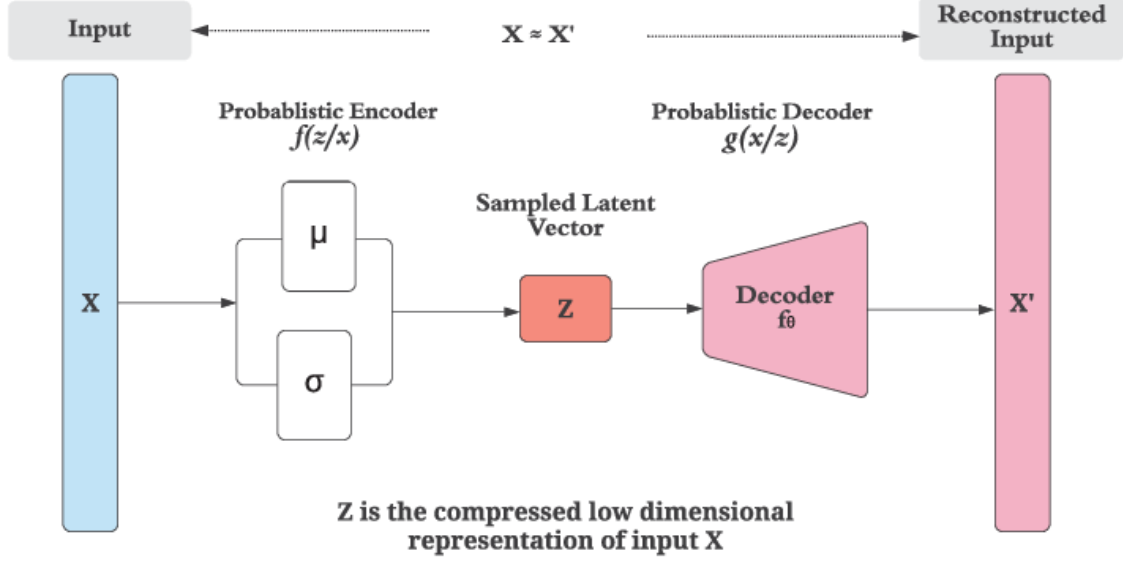


Figure 3.4: Architecture of Variational Autoencoder (VAE) to extract embeddings

3. Multi-Modal Feature Extraction

Depth Map Generation: In this work, we propose a Graph Convolutional Neural Network (GCNN)-based methodology to generate depth map representations from EEG signals. We have used results from MiDaS model (meant for Image to Depth conversion) as a ground truth for the GCNN network. The depth Maps which we get from the MiDaS model (from the corresponding Images of the EEG signal) is used as the ground truth for the GCNN network architecture to evaluate the performance of GCNN architecture. EEG data is first modeled as a graph where each electrode serves as a node and edges represent spatial or functional connectivity between electrodes. The raw time-series signals are used as node features, capturing the temporal dynamics of brain activity. The GCNN architecture, comprising multiple graph convolution layers followed by non-linear activations, dropout, and global pooling, extracts spatial-temporal features from this graph. These features are then passed through fully connected layers to predict a depth-related output, such as a scalar depth value or a compressed depth embedding.

The model is trained using regression loss functions like Mean Squared Error (MSE), and evaluated using metrics such as MAE, RMSE, and Cosine Similarity. These metrics measure the fidelity of the predicted depth with respect to ground-truth annotations or feature embeddings.

Basically the underlying hypothesis is that the spatial and functional relationships between EEG channels can be effectively modeled using graph structures, and that learning such representations can bridge neural activity and visual depth perception or image understanding.

This graph-based approach effectively leverages the structural and dynamic properties of EEG signals to infer visual depth, demonstrating a novel pathway for neural decoding in vision-related tasks.

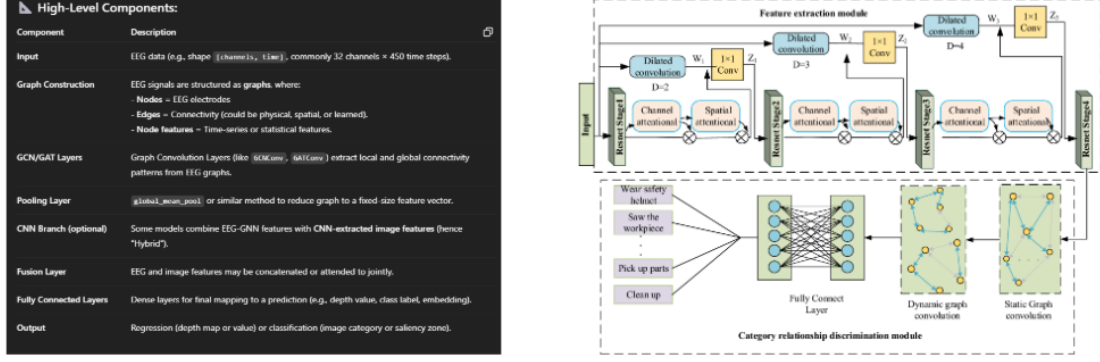


Figure 3.5: Architecture of GCNN along with the description of the GCNN layers used to train EEG to Depth

Caption Generation: The BLIP model generated textual descriptions of the images. These captions add semantic context, enriching the representation of visual information. BLIP: Bootstrapping Language Image Pre-training is used for unified vision-language understanding and generation. BLIP is a new VLP framework which enables a wider range of downstream tasks than existing methods.

Data perspective: most state-of-the-art methods (e.g., CLIP (Radford et al., 2021), ALBEF (Li et al., 2021a), SimVLM (Wang et al., 2021)) pre-train on image-text pairs collected from the web. Despite the performance gain obtained by scaling up the dataset, BLIP paper shows that the noisy web text is suboptimal for vision-language learning.

It introduces two contributions from the model and data perspective :

(a) **Multimodal mixture of Encoder-Decoder (MED):** a new model architecture for effective multi-task pre-training and flexible transfer learning. An MED can operate either as a unimodal encoder, or an image-grounded text encoder, or an image-grounded text decoder. The model is jointly pre-trained with three vision-language objectives: image text contrastive learning, image-text matching, and image conditioned language modeling.

(b) **Captioning and Filtering (CapFilt):** a new dataset bootstrapping method for learning from noisy image-text pairs. We finetune a pre-trained MED into two modules: a captioner to produce synthetic captions given web images, and a filter to remove noisy captions from both the original web texts and the synthetic texts.

BLIP achieves state-of-the-art performance on a wide range of vision-language tasks, including image-text retrieval, image captioning, visual question answering, visual reasoning, and visual dialog. We also achieve state-of-the-art zero-shot performance when directly transferring our models to two video-language tasks: text-to-video retrieval and videoQA.

Therefore we are using BLIP model, because of its improved accuracies and state of the art results to generate captions from the Corresponding Images of the EEG.

We have used BLIP in our pipeline and produced some results. With experimentation we found out that BLIP-2 gives us better results with our pipeline. Though heavy in computation it works well with our pipeline thus have used the same.

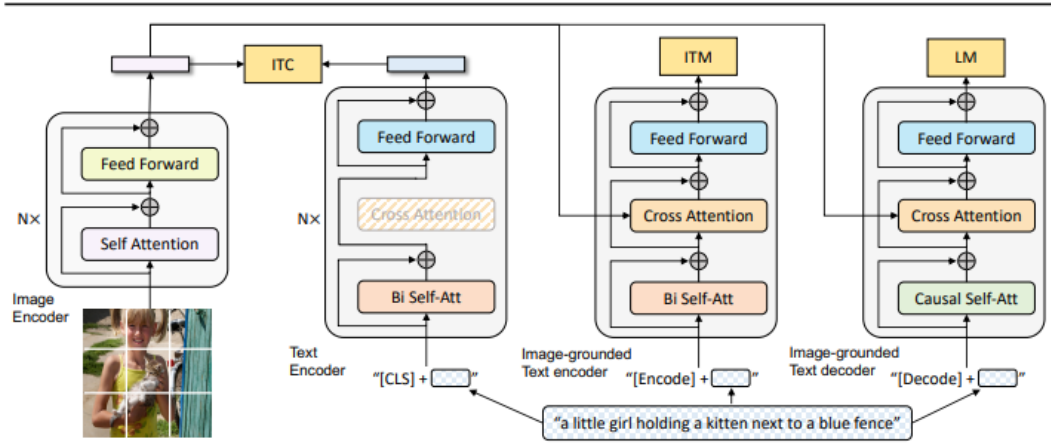


Figure 3.6: Pre-training model architecture and objectives of BLIP (same parameters have the same color). We propose multimodal mixture of encoder-decoder, a unified vision-language model which can operate in one of the three functionalities: (1) Unimodal encoder is trained with an image-text contrastive (ITC) loss to align the vision and language representations. (2) Image-grounded text encoder uses additional cross-attention layers to model vision-language interactions, and is trained with a image-text matching (ITM) loss to distinguish between positive and negative image-text pairs. (3) Image-grounded text decoder replaces the bi-directional self-attention layers with causal self-attention layers, and shares the same cross-attention layers and feed forward networks as the encoder. The decoder is trained with a language modeling (LM) loss to generate captions given images.

Further moving one step further we are using BLIP-2 which is an enhanced version of BLIP and the present state of the art model for Image. In our approach, we use BLIP-2 to generate scene-descriptive captions directly from images, leveraging its two-stage architecture. First, a frozen vision encoder (ViT-G) extracts high-level image features. These features are then passed through a trainable Q-former, which learns to produce a compact set of query embeddings that align with the language domain. Finally, these embeddings condition a frozen LLM (FlanT5) to generate coherent and semantically rich textual outputs. The model is trained with a combination of image-text matching and language modeling objectives, enabling robust generalization to various downstream tasks. In our pipeline, BLIP-2 serves as a powerful baseline or target text generator that captures depth-related semantics from images, which can later be compared or aligned with EEG-derived outputs.

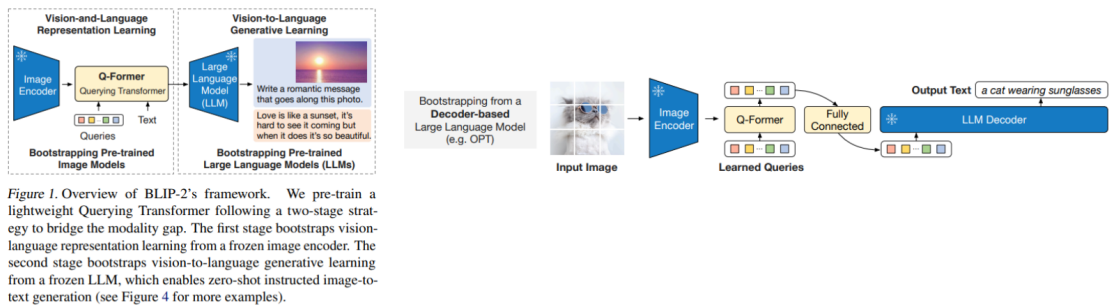


Figure 1. Overview of BLIP-2's framework. We pre-train a lightweight Querying Transformer following a two-stage strategy to bridge the modality gap. The first stage bootstraps vision-language representation learning from a frozen image encoder. The second stage bootstraps vision-to-language generative learning from a frozen LLM, which enables zero-shot instructed image-to-text generation (see Figure 4 for more examples).

Figure 3.7: BLIP-2 Architecture

4. Cross Modal Alignment

The EEG embeddings and the text embeddings are brought into a common latent space and both the embeddings are brought closer using the CLIP model. This alignment ensures that embeddings derived from EEG Signals correspond to meaningful textual features, facilitating effective multi-modal integration. We use contrastive learning optimized on cosine similarity matrix to convert EEG embeddings to relevant text embeddings this is bringing the EEG embeddings close to the Text Embeddings in the common latent space. In this way the EEG embeddings are converted into its corresponding Text embeddings.

Further to enhance the cosine similarity more, we use the concept of masking. One of the problems which we encounter with dealing with EEG signals is that EEG signal generated for different people is different for the same image shown, thus our cross modal alignment learning suffers and the cosine similarity reduces. To avoid this, we use the concept of masking. We have 10 subjects in our dataset and it has been captured as 16740 EEG samples combined for the 40 label classes images shown to all of these 10 subjects. Therefore fine-tuning of the CLIP incurs a low cosine similarity. Thus to improve it when we are doing cross modal training, this is optimizing on contrastive loss via cosine similarity matrix we mask the unmatched pair for each subject. By doing this the similarity of the matched pair this is alike pair increases and thus the conversion to relevant Text embedding becomes more efficient. This is the similarity for the alike pairs become more thus the generation of valid text through EEG embeddings also become more accurate. Therefore with masking of the unmatched pairs corresponding to different subjects the overall cosine similarity increases from 0.22 to 0.347, which depicts that the text embedding generated from the EEG embedding will be more precise for the Decoder to give a more descriptive and correct Text corresponding to the EEG signal.

Therefore by using Masking our overall accuracy of EEG to Text gets increased.

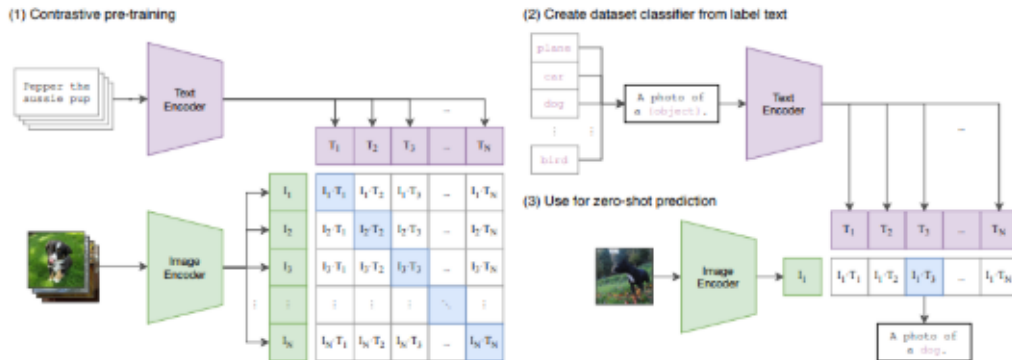


Figure 3.8: CLIP architecture to bring both EEG and Text Embeddings closer

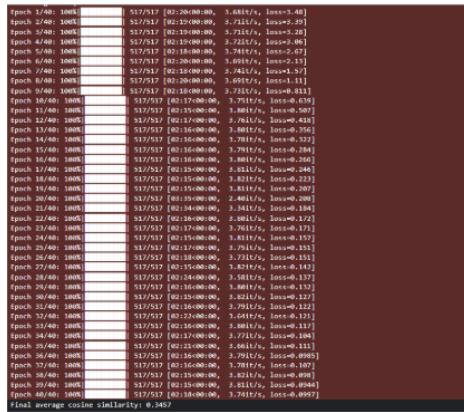


Fig : The training process of CLIP showing the contrastive loss reduce from 3.48 to 0.121 in 30 epochs showing that the text and the eeg embeddings are brought closer

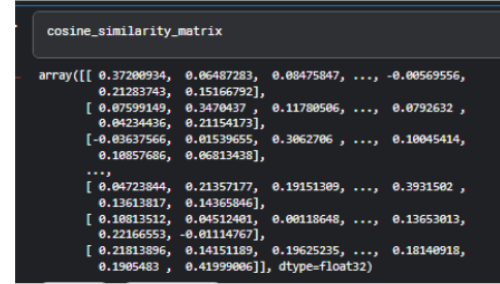


Fig : The final cosine similarity matrix depicting higher weight along the diagonals (matching pairs are given more weight than non matching pairs)
[Note: We are planning to use Masking in CLIP to improve the overall performance]

Figure 3.9: Figure shows the average loss of the CLIP decreasing(3.48 to 0.121) with every epochs indicating the EEG embeddings and the Text embeddings are brought closer and the cosine similarity matrix showing how the cross alignment is performing (0.35 average cosine similarity)

5. GPT-2 Text Decoder

These projected embeddings are fed into the GPT-2 language model, which then autoregressively generates coherent and contextually relevant text. During training, the model minimizes cross-entropy loss on the next-token prediction, while inference uses sampling strategies like top-k or nucleus sampling to produce diverse and meaningful text outputs. Thus after we get the Text Embedding from the EEG by cross modal alignment we fed them into GPT-2 Text Decoder which generate the corresponding text from EEG. Therefore in this way we convert EEG directly to Text.

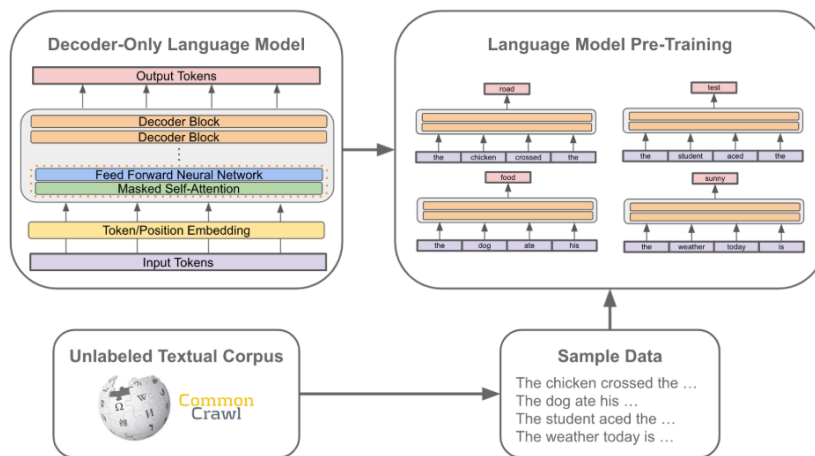


Figure 3.10: GPT-2 architecture

5. Image Reconstruction with Stable Diffusion

Stable Diffusion operates in a latent space, where it first encodes image representations using a variational autoencoder (VAE) and then performs denoising via a UNet-based denoising network conditioned on text embeddings derived from a frozen CLIP text encoder. In our setup, the text—either directly from BLIP-2 or reconstructed from EEG-derived features—is passed to the diffusion model to guide the image generation process. We divided the step of Stable Diffusion into two phases:

1. Experimental Phase: Multiple experiments are conducted to evaluate the reconstruction capabilities of the stable diffusion model and the combination of inputs which generate the best result from the stable diffusion model.

Noisy Image Input: Inputs include noisy images paired with text, labels, or empty prompts.

Depth Map Input: Depth maps are paired with text, labels, or empty prompts. These experiments reveal that BLIP2-generated captions, when used as prompts, produce the most accurate reconstructions.

2. Final Reconstruction Pipeline: The output from Cross Modal Alignment (EEG to Text) and the depth map from GCNN (Graph Convolution Neural Networks) (EEG to Depth) both combined as an input to the stable diffusion model produce the best results and reconstructions. We take the pre-trained stable diffusion model which takes depth map as input conditioned on text prompts and reconstruct the image. (Visual Decoding using EEG). In this way we are able to reconstruct image solely based on EEG signal.

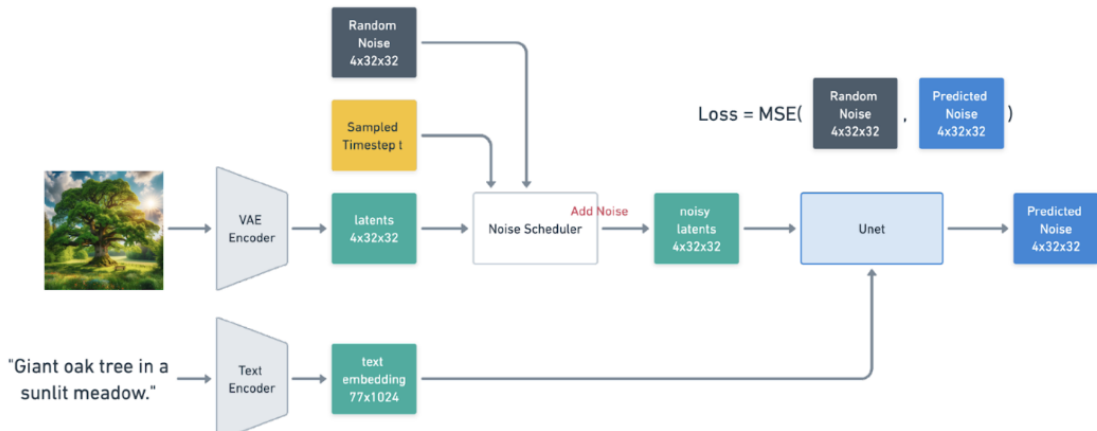


Figure 3.11: Stable diffusion methodology to reconstruct images

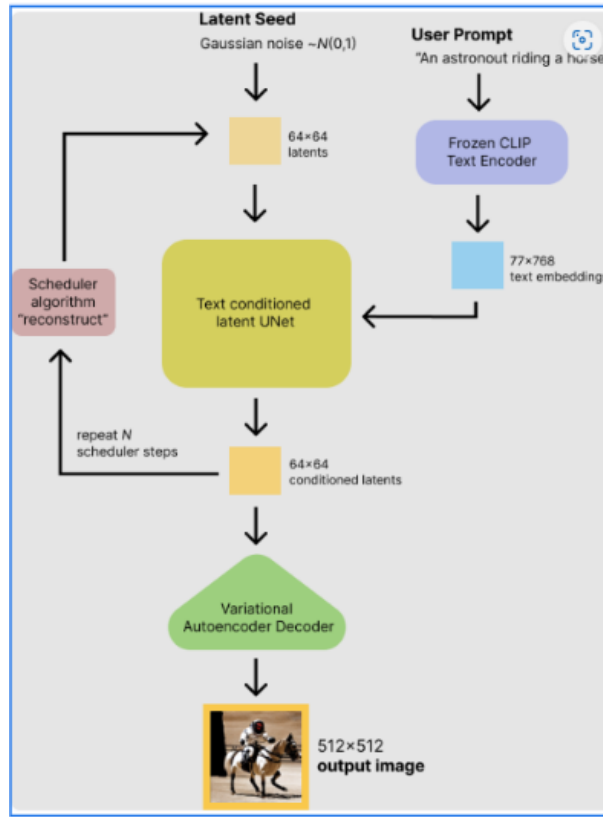


Figure 3.12: CompVis/val-2 stable diffusion model architecture to reconstruct the image

7. Evaluation metrics

Structural Similarity Index(SSIM): We use SSIM evaluation metric as a measure of the visual similarity between reconstructed and original images.

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)}$$

with:

- μ_x the **average** of x ;
- μ_y the **average** of y ;
- σ_x^2 the **variance** of x ;
- σ_y^2 the **variance** of y ;
- σ_{xy} the **covariance** of x and y ;
- $c_1 = (k_1 L)^2$, $c_2 = (k_2 L)^2$ two variables to stabilize the division with weak denominator;
- L the **dynamic range** of the pixel-values (typically this is $2^{\text{\#bits per pixel}} - 1$);
- $k_1 = 0.01$ and $k_2 = 0.03$ by default.

Figure 3.13: SSIM (Structural Similarity) Formula and Significance

4 Theoretical/Numerical/Experimental findings

Results from BLIP-2

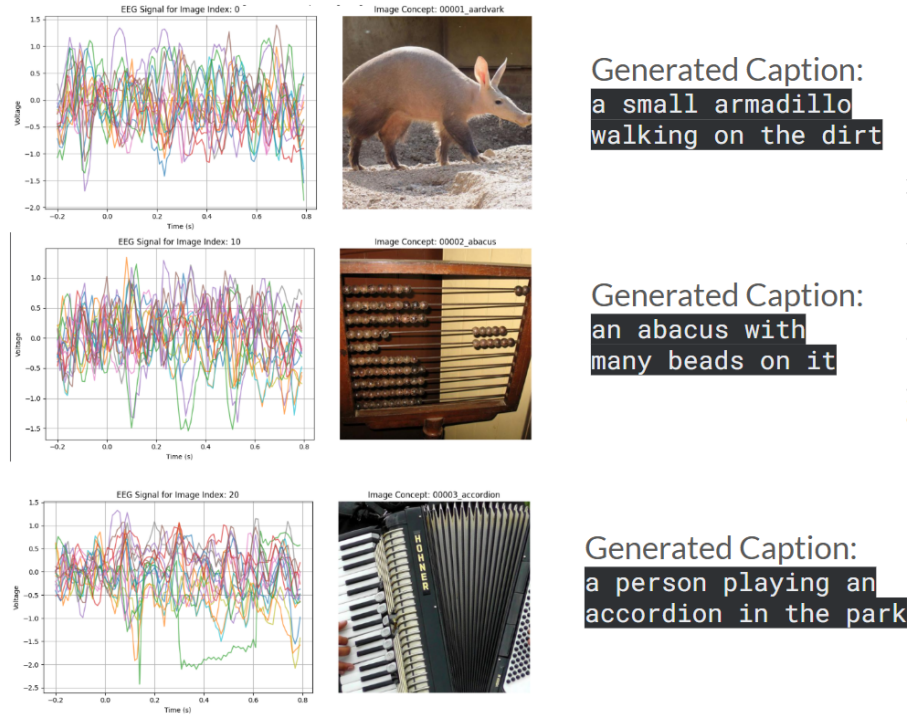


Figure 4.1: Results from BLIP-2 model generating captions from Images taking it as the ground truth for further processing

Models	#Trainable Params	NoCaps Zero-shot (validation set)								COCO Fine-tuned Karpathy test	
		in-domain		near-domain		out-domain		overall		B@4	C
		C	S	C	S	C	S	C	S		
OSCAR (Li et al., 2020)	345M	-	-	-	-	-	-	80.9	11.3	37.4	127.8
VinVL (Zhang et al., 2021)	345M	103.1	14.2	96.1	13.8	88.3	12.1	95.5	13.5	38.2	129.3
BLIP (Li et al., 2022)	446M	114.9	15.2	112.1	14.9	115.3	14.4	113.2	14.8	40.4	136.7
OFA (Wang et al., 2022a)	930M	-	-	-	-	-	-	-	-	43.9	<u>145.3</u>
Flamingo (Alayrac et al., 2022)	10.6B	-	-	-	-	-	-	-	-	-	138.1
SimVLM (Wang et al., 2021b)	~1.4B	113.7	-	110.9	-	115.2	-	112.2	-	40.6	143.3
BLIP-2 ViT-g OPT _{2.7B}	1.1B	<u>123.0</u>	<u>15.8</u>	117.8	<u>15.4</u>	123.4	15.1	119.7	<u>15.4</u>	<u>43.7</u>	145.8
BLIP-2 ViT-g OPT _{6.7B}	1.1B	123.7	<u>15.8</u>	<u>119.2</u>	15.3	<u>124.4</u>	14.8	<u>121.0</u>	15.3	43.5	145.2
BLIP-2 ViT-g FlanT5 _{XL}	1.1B	123.7	16.3	120.2	15.9	124.8	15.1	121.6	15.8	42.4	144.5

Table 3. Comparison with state-of-the-art image captioning methods on NoCaps and COCO Caption. All methods optimize the cross-entropy loss during finetuning. C: CIDEr, S: SPICE, B@4: BLEU@4.

Figure 4.2: Comparison of BLIP-2 with other models

Results of Cross Modal Alignment and GPT-2 Decoder (EEG to Text)

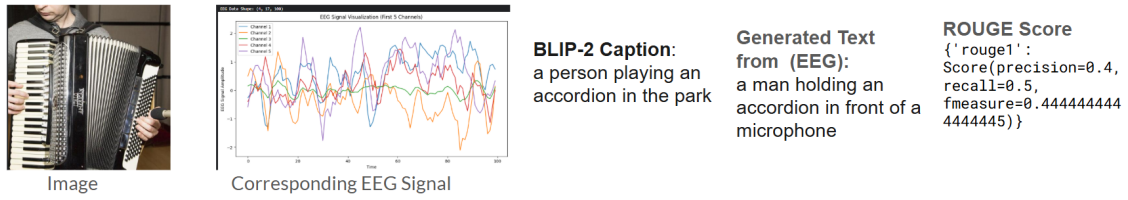


Figure 4.3: Example result 1 from EEG to Text (rouge1': Score(precision=0.40, recall=0.50, fmeasure=0.44))

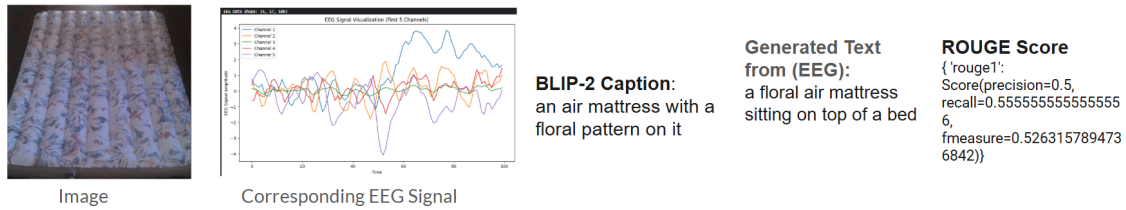


Figure 4.4: Example Result 2 from EEG to Text ('rouge1': Score(precision=0.50, recall=0.55, fmeasure=0.52))

Results of GCNN architecture (EEG to Depth)

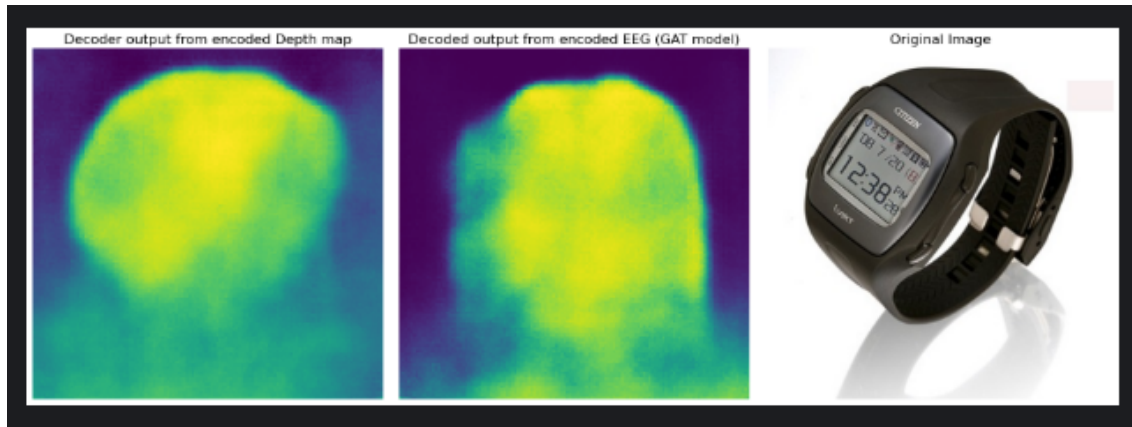


Figure 4.5: Example Result from EEG to Depth using GCNN architecture

Final Results of the entire pipeline this is EEG to Image (Visual Decoding using EEG)

SSIM : 11.02

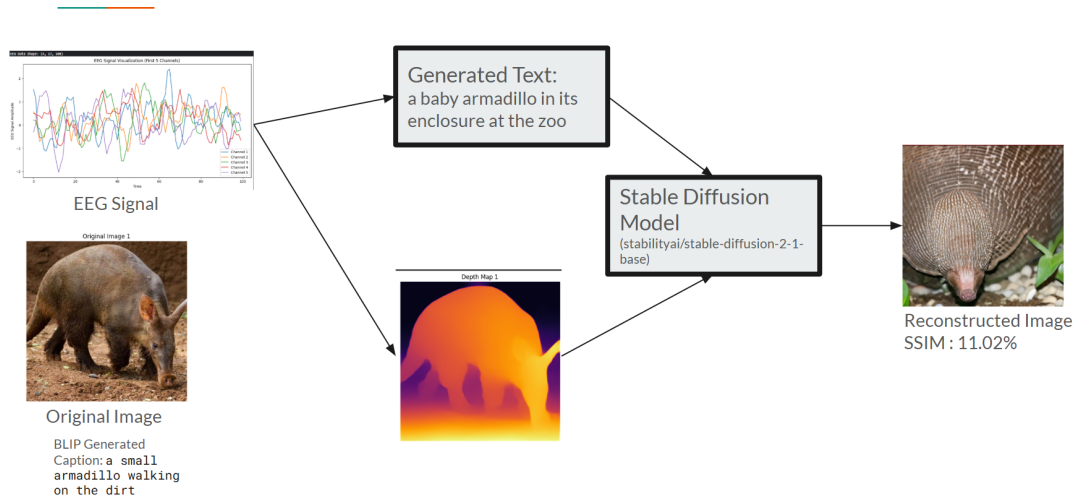


Figure 4.6: Example Result 1 of the final pipeline from stable diffusion (Image reconstruction) this is Visual Decoding using EEG (**SSIM: 11.02 percent**)

SSIM : 14.32

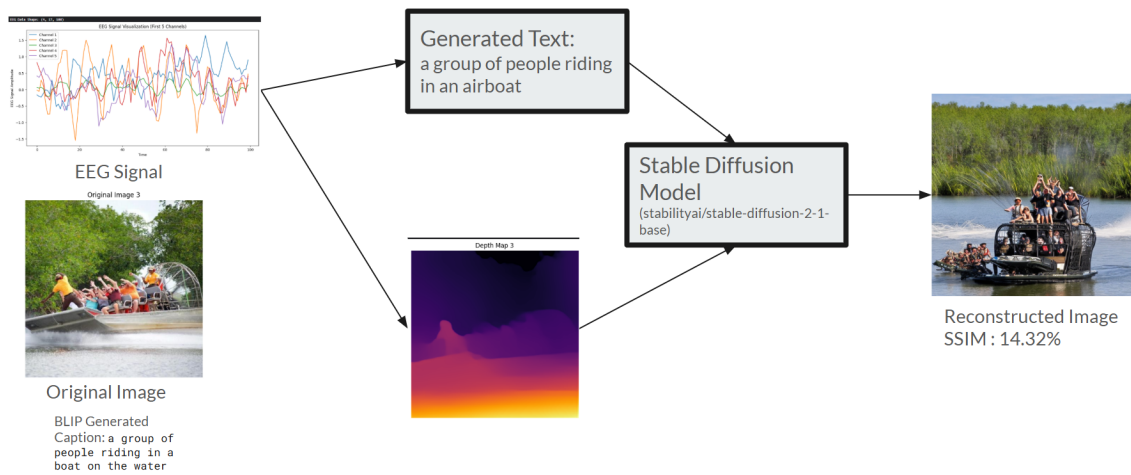


Figure 4.7: Example Result 2 of the final pipeline from stable diffusion (Image reconstruction) this is Visual Decoding using EEG (**SSIM: 14.32 percent**)

Quantitative Results and Ablation Study

S.No.	Noisy Image	Depth-Map	VAE	LABRAM	EEGNet	Transformers for direct EEG to Text Conversion	CLIP (without masking)	CLIP (with masking)	Prompt	Final Test SSIM
1	✓	✗	✓	✗	✗	✗	✓	✗	""	7.31
2	✓	✗	✓	✗	✗	✗	✓	✗	Class Label	8.55
3	✓	✗	✓	✗	✗	✗	✓	✗	BLIP Caption	11.23
4	✗	✓	✓	✗	✗	✗	✓	✗	""	9.56
5	✗	✓	✓	✗	✗	✗	✓	✗	Class Label	11.34
6	✗	✓	✓	✗	✗	✗	✓	✗	BLIP Caption	13.45
7	✗	✓	✗	✓	✗	✗	✓	✗	Generated Caption	15.15
8	✗	✓	✗	✗	✓	✗	✓	✗	Generated Caption	12.04
9	✗	✓	✓	✗	✗	✓	✗	✗	Generated Caption	10.29
10	✗	✓	✓	✗	✗	✗	✓	✗	Generated Caption	18.13
11	✗	✓	✓	✗	✗	✗	✗	✓	Generated Caption	24.14

Figure 4.8: Quantitative Results and the Ablation study showing the experimentation results on testing data and our final test SSIM

To evaluate the impact of different modules and input types on the performance of EEG-to-image reconstruction, we conducted a comprehensive ablation study across eleven experimental configurations. Each row in the table represents a unique setup combining various components such as Noisy Image input, Depth Map usage, VAE, LABRAM, EEGNet, and different encoding schemes for EEG-to-text transformation. We also tested the effect of different textual prompts fed into the CLIP and Stable Diffusion pipelines, ranging from simple class labels to captions generated by BLIP-2 or transformers trained directly on EEG signals. The Structural Similarity Index (SSIM) was used as the final evaluation metric to quantify image quality. Initial experiments using only noisy images and basic prompts yielded lower SSIM scores (e.g., 7.31). Performance steadily improved with the incorporation of more semantic input like depth maps and generated captions. The highest SSIM score of 24.14 was achieved in configuration 11, which utilized a depth map along with a VAE and generated captions derived from EEG-based textual decoding, showing the superiority of combining depth-aware features with semantically rich, EEG-driven prompts. This systematic comparison highlights the critical role of multi-modal fusion and neural captioning in achieving high-fidelity reconstructions from brain signals.

5 Summary and Conclusion

This project set out to explore how we can turn brain activity, recorded through EEG, into meaningful and accurate images. We built a pipeline that brought together different techniques—from understanding depth through EEG signals, to generating text descriptions of images using BLIP-2, and even translating EEG into text using a specially trained masked CLIP model.

Our approach centered on breaking down the problem into smaller, more manageable pipelines, each contributing a specific piece to the final image reconstruction.

- **Depth Map Extraction from EEG:** We started by using EEG signals to extract depth information. This required interpreting spatial and temporal patterns in the brainwaves to form a sense of scene structure—something not typically straightforward using EEG alone.
- **Caption Generation using BLIP-2:** To bring in a textual understanding of what was in the image, we used BLIP-2 to generate captions. These captions helped add meaningful descriptions that guided the final image generation process.
- **Masked CLIP Model for EEG-to-Text Conversion:** Instead of relying solely on pre-existing captions, we also trained a masked CLIP model to convert EEG signals directly into textual embeddings. This allowed us to translate raw brain activity into semantic text representations.
- **Stable Diffusion for Image Reconstruction:** Finally, we combined all the components—a noisy base image, the generated text (either from BLIP-2 or EEG), and the depth map—into a Stable Diffusion model. This fusion allowed the system to reconstruct visuals that aligned with the brain’s activity.

Along the way, we experimented with different combinations of these components to see what worked best. We learned that both depth information and strong textual prompts (especially those generated from EEG) played a big role in improving the quality of the final image. Our evaluation, primarily through SSIM scores, confirmed this.

Notably, the current benchmark accuracy for image reconstruction from EEG stands at approximately **34.4%**, which highlights the complexity of the task and also offers a valuable reference point for measuring future improvements.

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